

ECE275 Sample exam Spring 2026

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Student Name:

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1 Instructions

- There are seven problems. All problems are required.
- Maximum number of marks is 100. This exam amounts 10% toward the final grade.
- Time allowed is 120 minutes.
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8×11 in two-sided cheatsheet is allowed.
- Non-programmable calculators are permitted.
- Please use a pen or heavy pencil to ensure legibility. Colored pens/pencils are recommended for K-map grouping.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

Problem 1. *Simplify the following Boolean equation using Boolean theorems. Check for correctness using a K-map (5 marks).*

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}C\bar{D} + ABD + \bar{A}\bar{B}C\bar{D} + B\bar{C}D + \bar{A}$$

- Problem 2.** Implement the function $Y = BC + \bar{A}\bar{B}\bar{C} + B\bar{C}$ using
- (a) an 8:1 multiplexer (5 marks)
 - (b) a 4:1 multiplexer and no other gates (5 marks)
 - (c) a 2:1 multiplexer, one OR gate, and an inverter (5 marks)

Problem 3. Your company, Detect-o-rama, would like to design an Finite State Machine (FSM) that takes two inputs, A and B , and generates one output, Z . The output in cycle n , Z_n , is either the Boolean AND or OR of the corresponding input A_n and the previous input A_{n-1} , depending on the other input, B_n :

$$Z_n = A_n A_{n-1} \quad \text{if } B_n = 0 \quad (1)$$

$$Z_n = A_n + A_{n-1} \quad \text{if } B_n = 1 \quad (2)$$

(a) Is this FSM a Moore or a Mealy machine? (2 marks)

(b) Draw the state transition table (18 marks).

Use the following header for the state table

Present State	Next State				Outputs (Z_n)			
	Inputs $A_n B_n = 00$	01	10	11	$A_n B_n = 00$	01	10	11

Problem 4. A sequential circuit is to be used to control the operation of a vending machine which dispenses a \$0.25 product. The circuit has three inputs (N, D, and Q) and two outputs (R and C). The coin detector mechanism in the vending machine is synchronized with the same clock as the sequential circuit you are to design. The coin detector outputs a single 1 to the N, D, or Q input for every nickel (5 cents), dime (10 cents), or quarter (25 cents), respectively, that the customer inserts. Only one input will be 1 at a time. When the customer has inserted at least \$0.25 in any combination of nickels, dimes, and quarters, the vending machine must give change and dispense the product. The coin return mechanism gives change by returning nickels to the customer. For every 1 output on C, the coin return mechanism will return one nickel to the customer. The product is dispensed when the circuit outputs a single 1 on output R. The circuit should reset after dispensing the product.

Example: The customer inserts a nickel, a dime, and a quarter. The circuit inputs and outputs could look like this:

Inputs:

N=	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
D=	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
Q=	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
R=	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
C=	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0

Note that any number of 0's can occur between 1 inputs. Complete the following Moore state table for the sequential circuit, and for each state indicate how much money the customer has inserted or how much change is due. Use the following header for the state table (PS means Present state)

Money inserted	Change due	PS	Next State				Output (RC)
			NDQ = 000	NDQ = 100	NDQ = 010	NDQ = 001	
0	0	S_0	S_0	S_1	S_2	S_3	00

(30 marks)

Problem 5. (30 marks) Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:

1. There is no direct connection from Rome to Trento
2. There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.
3. Regional trains connect towns in the same region
4. Intercity trains don't stop in small towns.

Problem 6. Find the expression for J_1 and K_1 assuming that J_1 and K_1 are inputs to the J-K flip flop that capture the state of the second most significant bit Q_1 of the following state encoded table. The state encoding table given with state encoding denoted as $Q_2Q_1Q_0$. Here X is the input and Z is the output. (20 marks).

Present State $Q_2Q_1Q_0$	Next State		Output	
	$X = 0$ $Q_2^+Q_1^+Q_0^+$	$X = 1$ $Q_2^+Q_1^+Q_0^+$	$X = 0$ Z	$X = 1$ Z
000	100	101	1	0
001	100	101	0	1
010	000	000	1	0
011	000	000	0	1
100	111	110	1	0
101	110	110	0	1
110	011	010	1	0
111	011	011	0	1